

ESP 32 CONNECTED TO DHT22 TEMPERATURE AND HUMIDITY SENSOR

Stage	Component Name	Designator(s)	Functional Flow Description
1. Power Entry	USB Type-C 16P	U4	Receives 5V (VBUS) from the computer via the physical USB-C cable.
2. USB Handshake	5.1kΩ Resistors	R7, R8	These resistors on the CC pins tell the computer to safely send 5V to the board.
3. Power Filtering	100nF Capacitors	C2, C3	These filter out high-frequency noise from the USB power line before it reaches the rest of the board.
4. Power Check	50Ω Resistor & LED	R1 & LED	Power flows through the resistor (R1) to light up the Power LED, confirming the board is getting 5V.
5. Voltage Regulation	AMS1117-3.3	U3	Steps down the 5V to a stable 3.3V required for the ESP32 and other components.
6. The "Brain"	ESP-WROOM-32E	U1 / U5	Receives 3.3V and becomes the central control hub for all sensing and data processing.
7. Logic Stability	10kΩ Resistors	R9, R10	Holds the EN (Reset) and IO0 (Boot) pins at 3.3V to keep the ESP32 from rebooting randomly.
8. Communication Bridge	CH340C	U8	Acts as the translator, converting USB data into Serial (TX/RX) messages the ESP32 can understand.
9. Programming Switch	Transistors	Q1, Q2	Automatically "presses" the Reset and Boot buttons for you when you upload code from your computer.
10. Environmental Sensing	DHT22 Sensor	U2	Measures temperature and humidity and sends it as digital pulses to Pin IO4 on the ESP32.

11. Signal Integrity	10kΩ Resistor	R11	Acts as a pull-up on the DHT22 data line, ensuring the signal is clean enough for the ESP32 to read.
12. User Indicator	50Ω Resistor & LED	R6 & LED	The ESP32 sends a signal through the resistor (R6) to blink the User LED to show the sensor is active.

Resistors in series

Component Name	Designator	Configuration	Purpose
20 ohms	R1	Series	Current limit for Power LED
30 ohms	R2	Series	Current limit for User LED
30 ohms	R3	Series	Impedance matching for USB lines

Resistors in Parallel

Component Name	Designator	Configuration	Purpose
20 ohms	R4	Series	Current limit for Power LED
100 ohms	R5	Series	Current limit for User LED
50 ohms	R6	Series	Impedance matching for USB lines